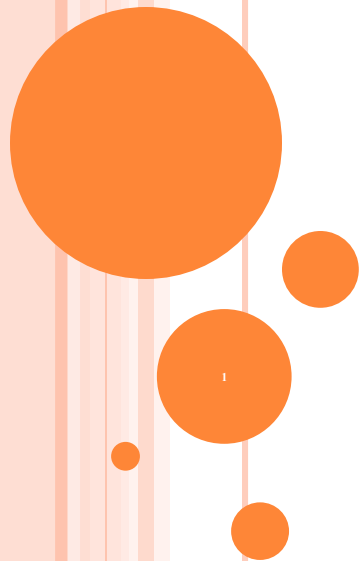


Wavelength Division Multiplexing (WDM) (cont.)

Salwa El-Sabban



WDM

WDM channel can be full duplex, half duplex, or simplex.

Full duplex

The communication between sender and receiver can occur simultaneously. The sender and receiver can both transmit and receive at the same time.

Half duplex

The communication between sender and receiver occurs in both directions, but only one at a time. The sender and receiver can both send and receive the information, but only one is allowed to send at any given time.

Simplex

The communication between sender and receiver occurs in only one direction. The sender can only send the data, and receiver can only receive the data. The receiver cannot reply to the sender.

WDM

For a two fiber full duplex WDM system, the two laser signals (one at both ends of the fiber link on different fibers) are assigned the same wavelength, which is referred to as the channel assignment.

However, in cases where only one fiber is used for full duplex communications, both lasers will have different wavelength assignments and therefore communication channel uses two WDM channels.

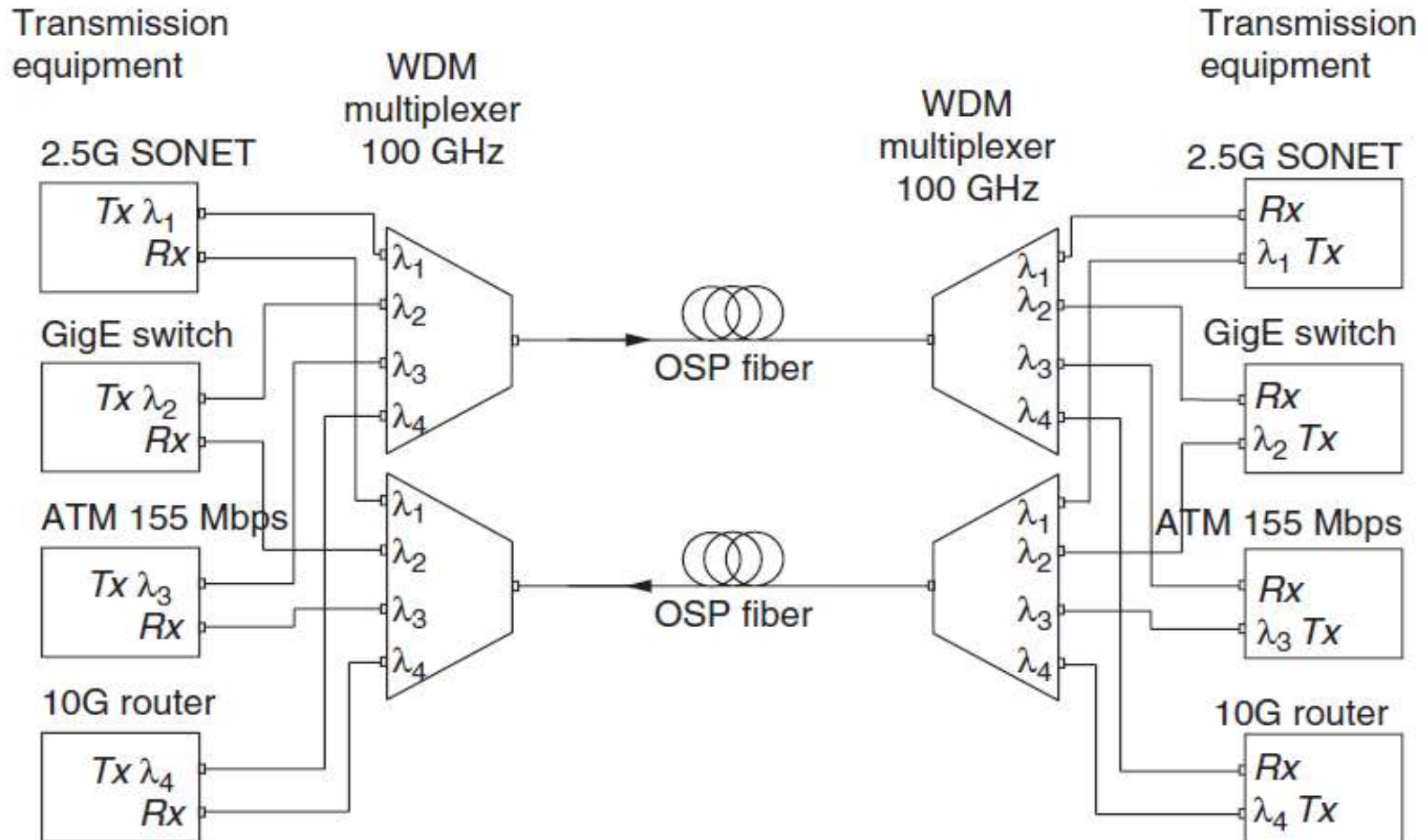
Typically, most WDM systems will use two fibers for a WDM link. Therefore, the WDM wavelength assignment is the same for both lasers for that channel.

WDM

A simple WDM system consists of WDM multiplexers located at each end of a fiber link. The channel side of the WDM, also known as the drop side, connects to the transmission equipment such as SONET gear, routers, and switches that are fitted with the proper WDM transceivers. The WDM transceiver must have both the **same channel assignment** as the multiplexer port that it will connect to, and the **same channel spacing** as the multiplexer. The common side of the multiplexer connects to the common fiber (aggregate fiber).

*Wavelength converters, referred to as transponders can be used to convert non-WDM optical signals (with standard standard 1310 or 1550 nm) to proper WDM signals. The transponders are inserted between the network equipment transceivers and WDM multiplexer.

WDM



Typical basic 4-channel WDM link.

WDM

WDM systems are available as simple(passive) or active systems.

The active systems are a combination of simple WDM's and some electronics that provide a rich set of features.

Typically, a simple system contains no electronics, and therefore has no power or cooling requirements. They are generally limited by the WDM transceiver's optical budget to spans of less than 100 km (assuming equipment optical budget of 28 dB, fiber loss of 0.22 dB/km, and multiplexer pair insertion loss of 6 dB).

Basic WDM(MUX) Types

1-Unidirectional

2-Bidirectional

3-Universal

4-Optical add/drop multiplexer (OADM)

WDM

Basic WDM(MUX) Types

1-Unidirectional

-The unidirectional WDM allows transmission in the WDM in only one direction, channel ports to common port or common port to channel ports.

-The channel port(s) in a unidirectional multiplexer (mux) unit are designated as In port(s) and connect to the equipment laser(s).

-The channel port(s) in a unidirectional demultiplexer (demux) unit are designated as Out port(s) and connect to the receiver(s) equipment.

-A laser cannot be connected to the demultiplexer unit and a receiver fiber cannot be connected to the multiplexer unit.

WDM

Basic WDM(MUX) Types

1-Unidirectional

-Unidirectional WDMs are always installed as a mux and demux pair with common ports connecting them by a single OSP fiber.

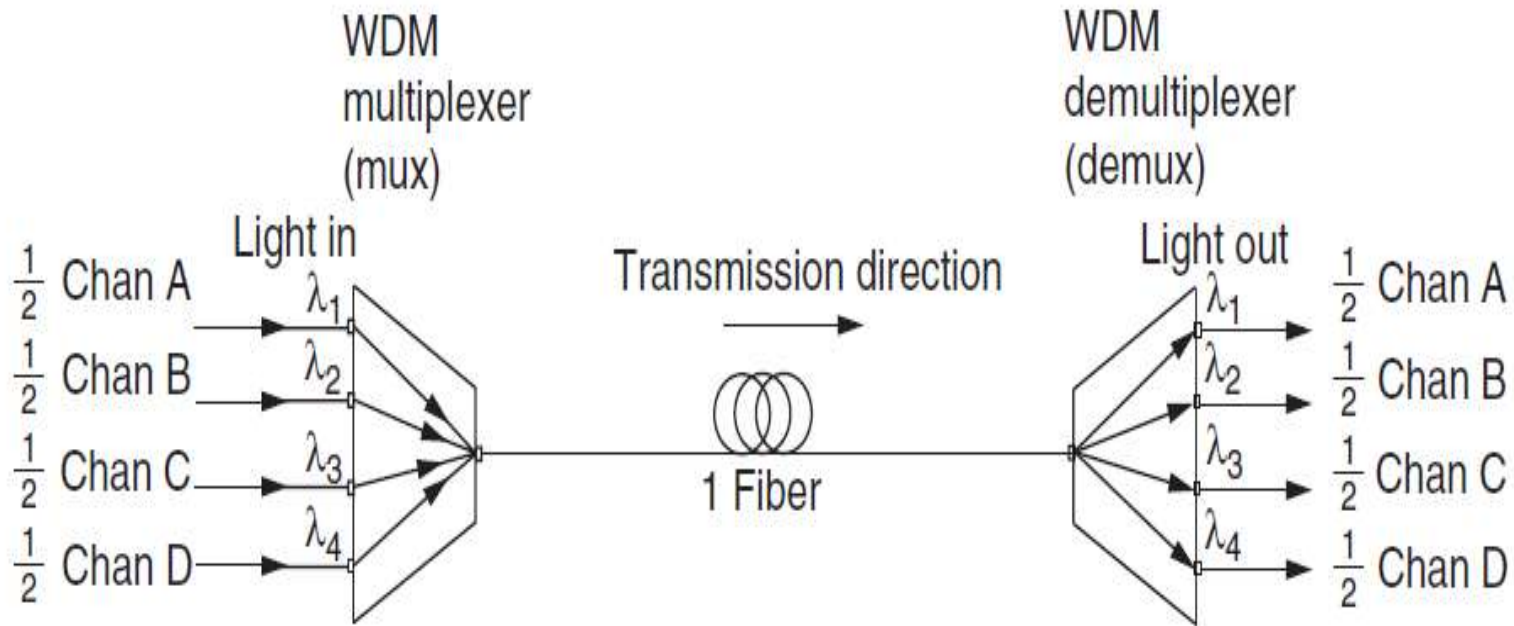
-Unidirectional WDM positions are not interchangeable

-Unidirectional WDMs are always used in two-fiber links for full duplex communications. Therefore, the WDM configuration shown in the following figure needs to be duplicated but in the opposite direction in order to produce a full duplex, 4-channel WDM system. This is a common WDM type deployed with EDFAs because they are also unidirectional.

WDM

Basic WDM(MUX) Types

1-Unidirectional



Unidirectional 4λ WDM

Basic WDM(MUX) Types

1-Unidirectional- dual unidirectional mux/demux unit

-Here both multiplexer and demultiplexer are combined into one unit at each end of the fiber. This system uses two common OSP fibers for the link connection.

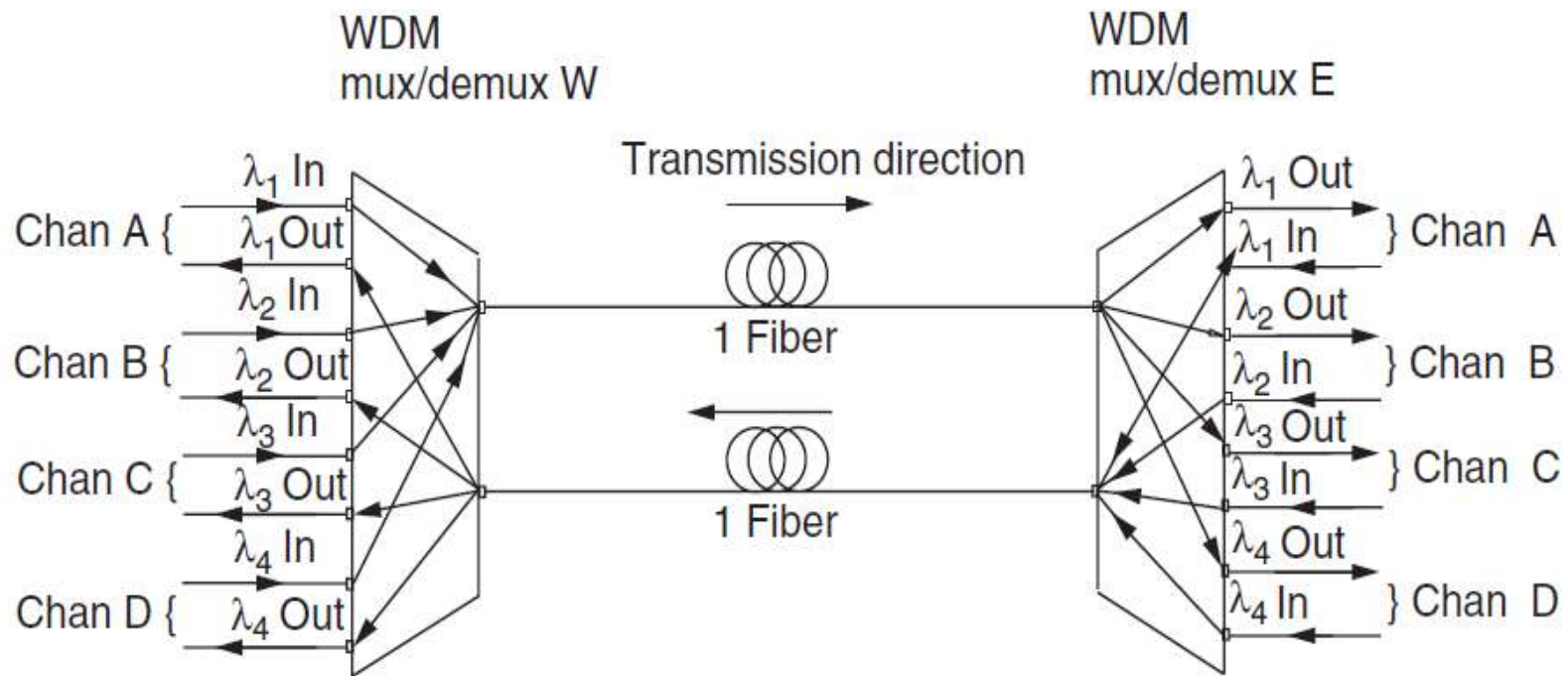
-Each fiber can carry transmission in only one direction.

-The units are referred to as Mux/Demux East and Mux/Demux West

WDM

Basic WDM(MUX) Types

1-Unidirectional- dual unidirectional mux/demux unit



Unidirectional 4-channel mux/demux WDM

WDM

Basic WDM(MUX) Types

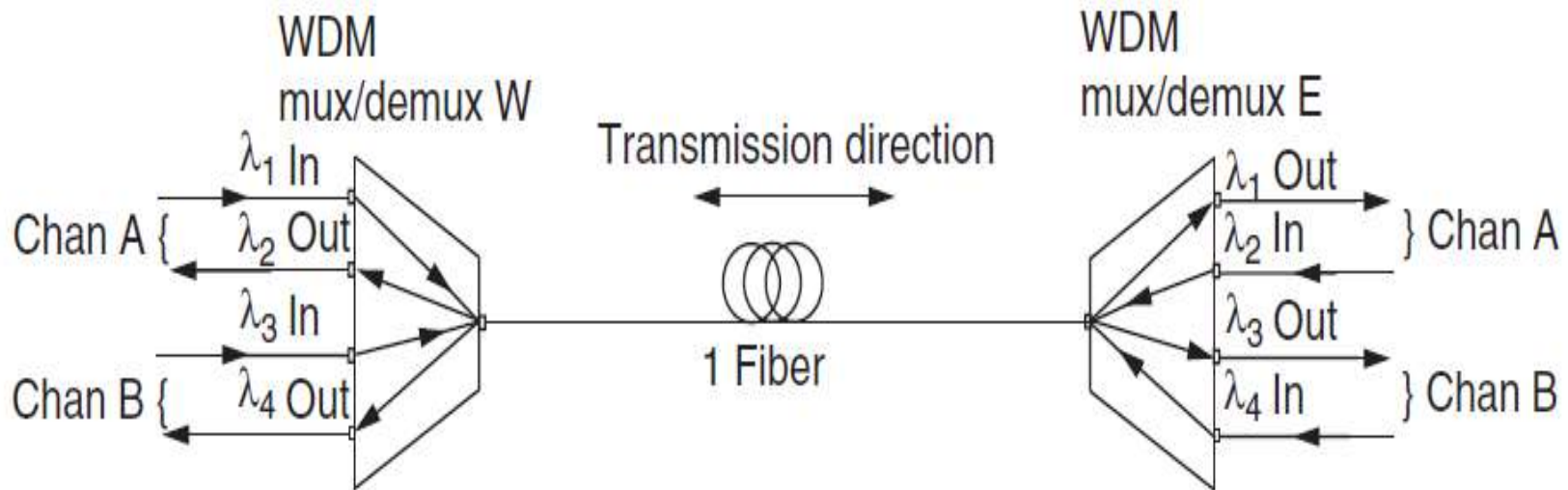
2-Bidirectional

- The bidirectional WDM allows transmission in both directions but laser and receiver equipment must connect to the appropriate In and Out channel ports.
- The common port is bidirectional. The units are referred to as Mux/Demux East and Mux/Demux West.
- Bidirectional WDM positions are not interchangeable

WDM

Basic WDM(MUX) Types

2-Bidirectional



Bidirectional 4λ WDM

WDM

Basic WDM(MUX) Types

3-Universal

-The universal WDM allows transmission in both directions on any channel ports and on the common port.

-The units can operate as both mux or demux and can be used at either end of the fiber link. This is a convenient feature that reduces complexity during design, ordering, and installation.

WDM

Basic WDM(MUX) Types

3-Universal

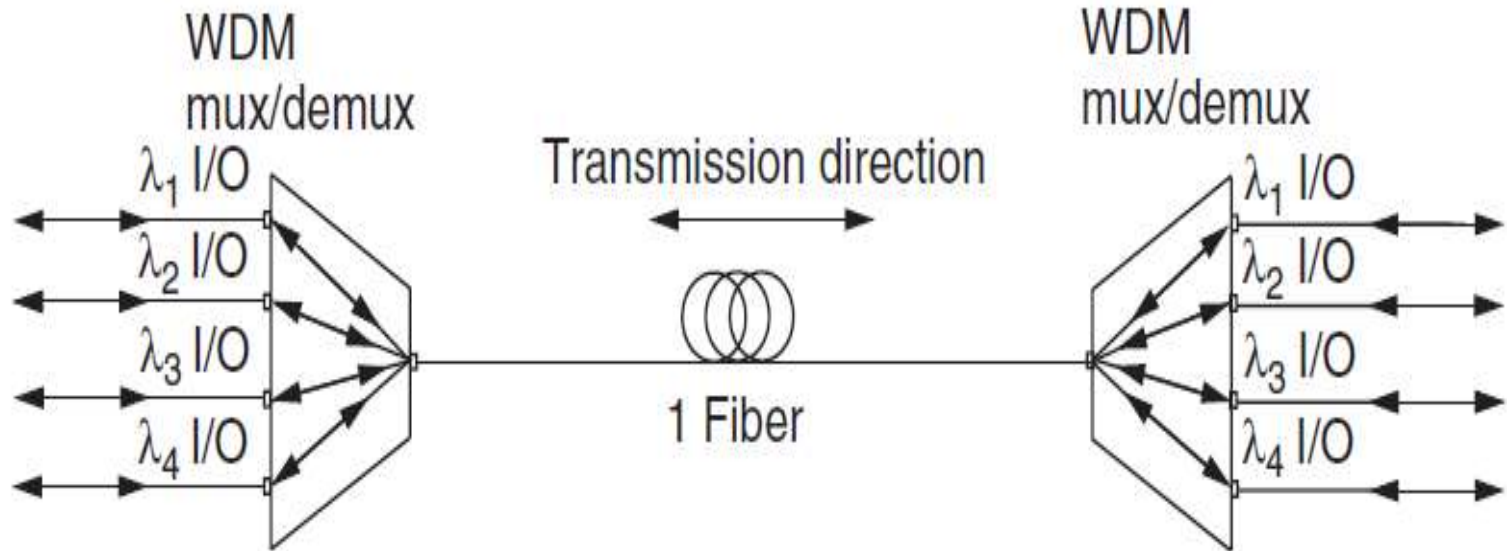
-It can be used at either end of the link. Although these units can be used in one-fiber links, they are typically used in two-fiber links.

-Optical amplifiers cannot be used for single-fiber bidirectional WDM transmission systems because most amplifiers are unidirectional. Therefore, for long-haul single-fiber applications, unidirectional WDMs need to be used.

WDM

Basic WDM(MUX) Types

3-Universal



Universal 4λ WDM

WDM

Basic WDM(MUX) Types

-The total number of bidirectional transmission channels depends on the number of channels the WDM can multiplex and the number of fibers that will be used in the transmission link.

If one fiber is used, then the total number of transmission channels is half of the total number of the WDM's channels (assuming full duplex communication).

If two fibers are used, the number of channels equals the number of WDM channels.

WDM

Basic WDM(MUX) Types

For full duplex (bidirectional) communication, two simultaneous transmission paths are required, one in each direction.

If two fibers are used in a link, then each fiber is designated a transmission direction and the two fibers are referred to as the transmit fiber and receive fiber.

To accomplish full duplex communication in one fiber, the WDM wavelengths are divided in half to determine the number of transmission channels. Each half represents a transmission direction in the fiber. This is because two unique wavelengths are required in one fiber, one in each direction, for one full duplex channel of communication

WDM

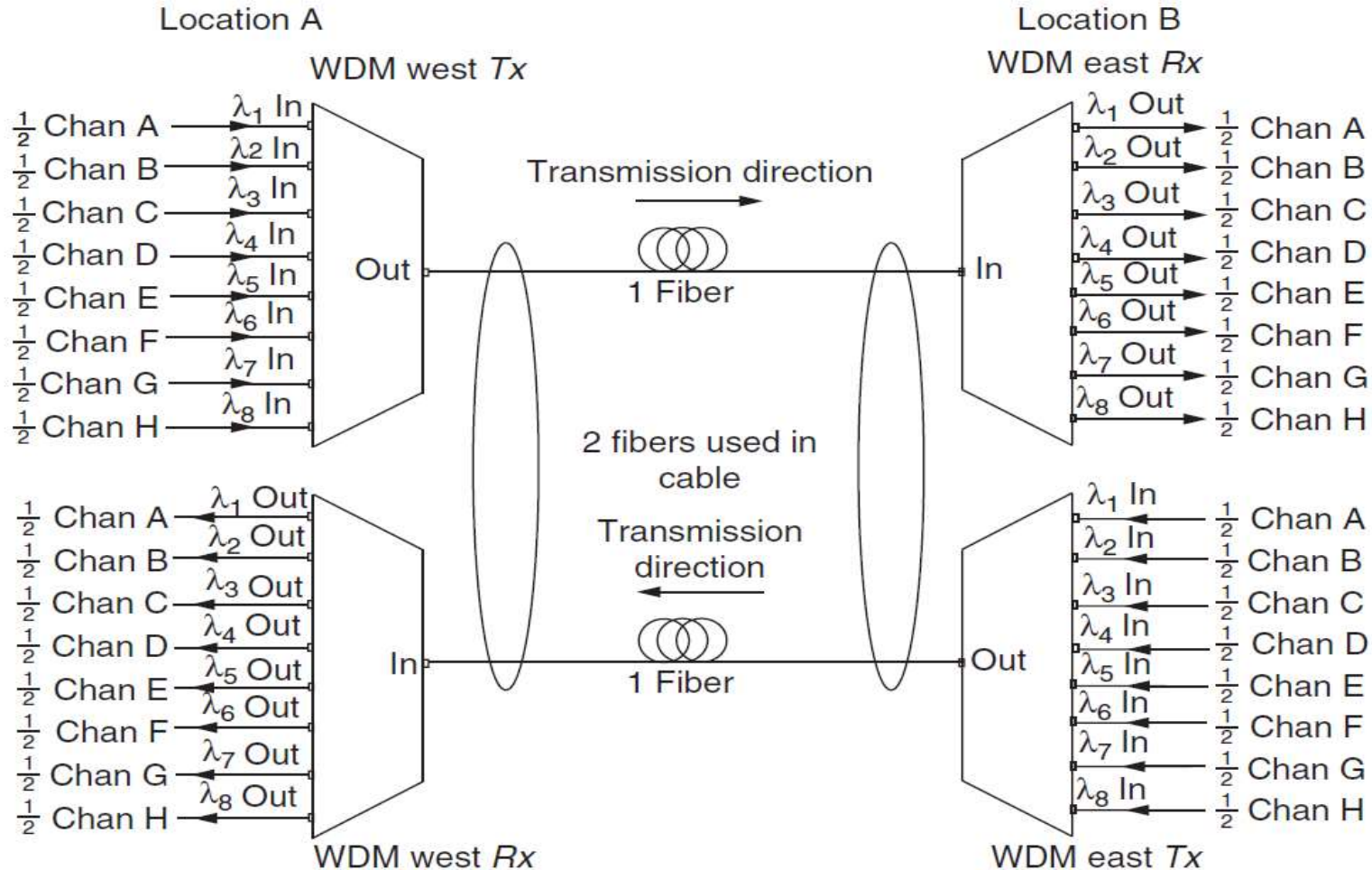
Basic WDM(MUX) Types

An 8-channel universal WDM can support only four full duplex transmission channels on a one-fiber link. Two of these WDM 8-channel units are required, one at each end of the one-fiber link.

However, if the same 8-channel universal DWDM is used in a two-fiber link, then it will support eight full duplex transmission channels and four WDM units are required for the link

WDM

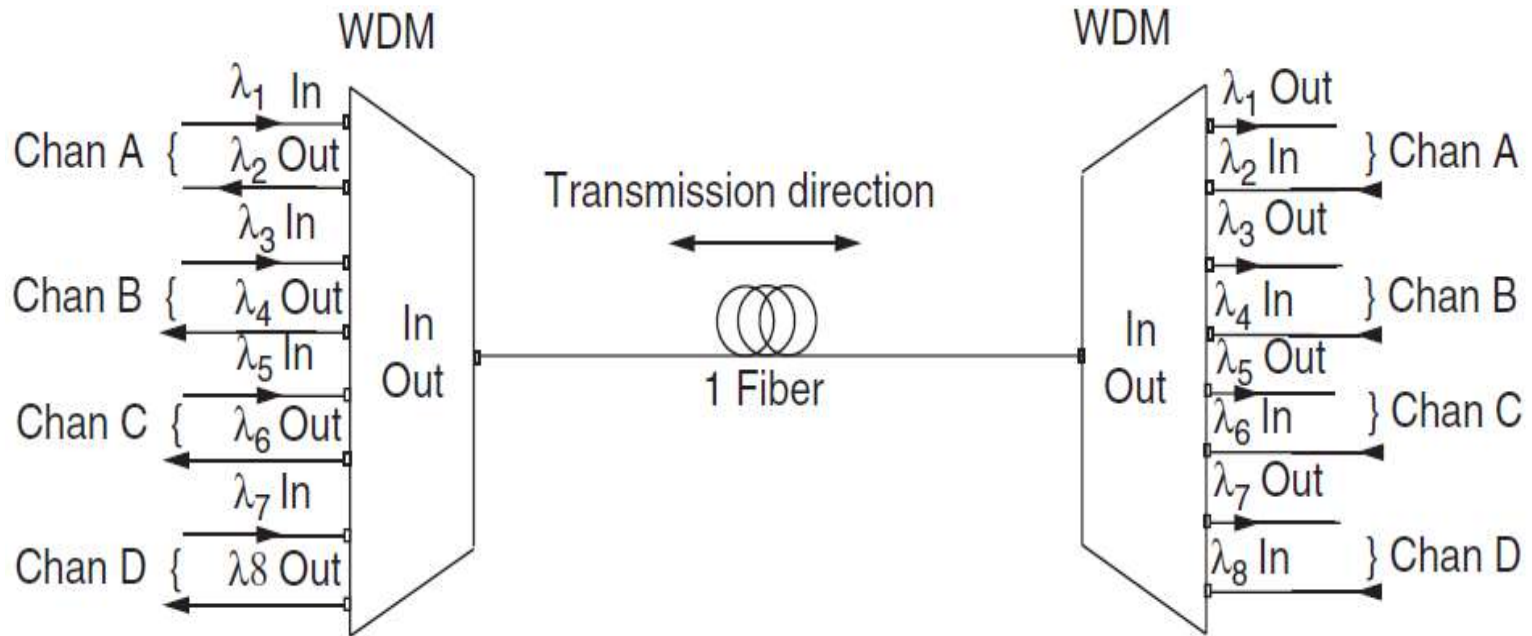
Basic WDM(MUX) Types



8-channel two-fiber WDM system

WDM

Basic WDM(MUX) Types



4-channel single-fiber WDM system

WDM

Basic WDM(MUX) Types

4-Optical add/drop multiplexer (OADM)

The OADM is a multiplexer that is designed to drop (and/or add) specific WDM channels in the middle of a fiber link. All other WDM channels pass through the unit.

The OADM has two common fiber ports, In and Out, for single-fiber OADM and four common fiber ports, In East, Out East, In West, and Out West for a two-fiber OADM, as well as drop ports.

OADM function can also be accomplished with back to back WDMs; however, the through channels loss will be much higher. An OADM is also typically less pricey than back to back WDMs, thereby reducing equipment costs.

WDM

Basic WDM(MUX) Types

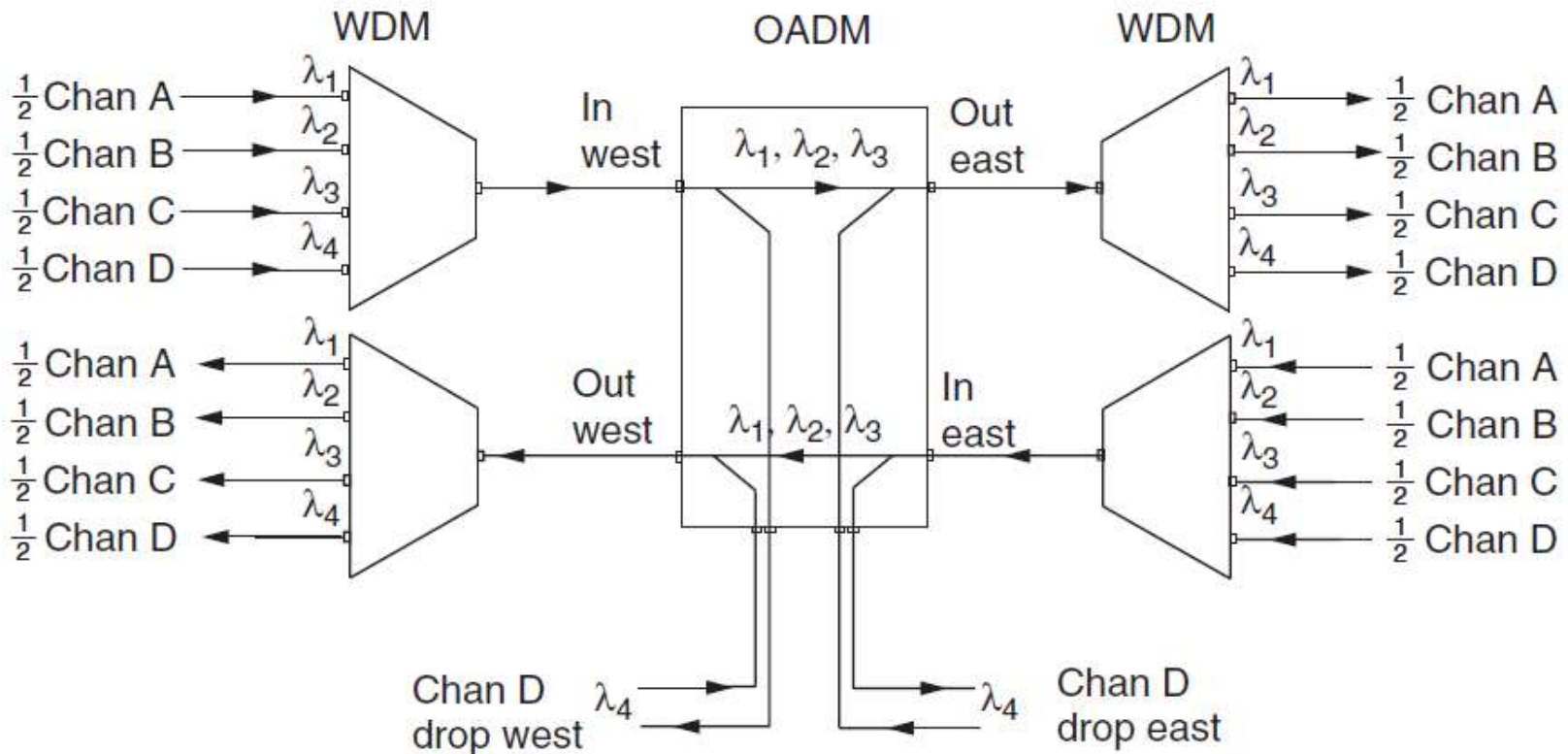
4-Optical add/drop multiplexer (OADM)

OADMs must match the WDM specifications of channel spacing and assignments in the link where they will be used. CWDM OADMs are deployed in CWDM links, and DWDM OADMs are deployed in DWDM links. DWDM OADMs must also match DWDM link channel spacing (50, 100, 200 GHz).

WDM

Basic WDM(MUX) Types

4-Optical add/drop multiplexer (OADM)



OADM dropping channel D east and west

WDM

Basic WDM(MUX) Types

4-Optical add/drop multiplexer (OADM)

